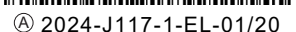


270 marks



Instructions

There are 13 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 (Suggested maximum time: 5 minutes)

Question 1 (Suggested maximum time: 5 minutes)

- (a)** Find the value of each of the following.

(i) $114 + 212$

[illegible]

(ii) $(6 + 4) \times 7$

[illegible]

(iii) 9×12.1

[illegible]

- (b)** Fill in the missing number in each of the following sequences:

(i)

2	4	6	
---	---	---	--

(ii)

2		-2	-4
---	--	----	----

(iii)

4	9		25
---	---	--	----

[illegible]

- (c) Write the numbers **3** and **16** into the two empty boxes below to make the mathematical statement true. Use each number only once.

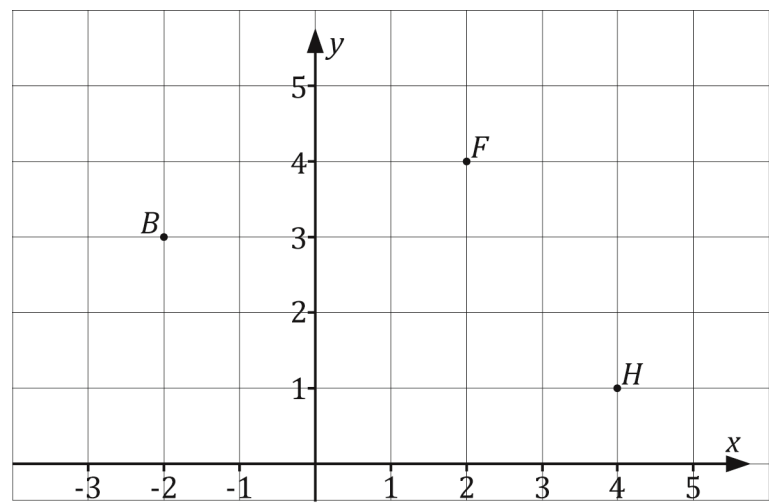
	9		15
8		=	16

[illegible]

Question 2

(Suggested maximum time: 10 minutes)

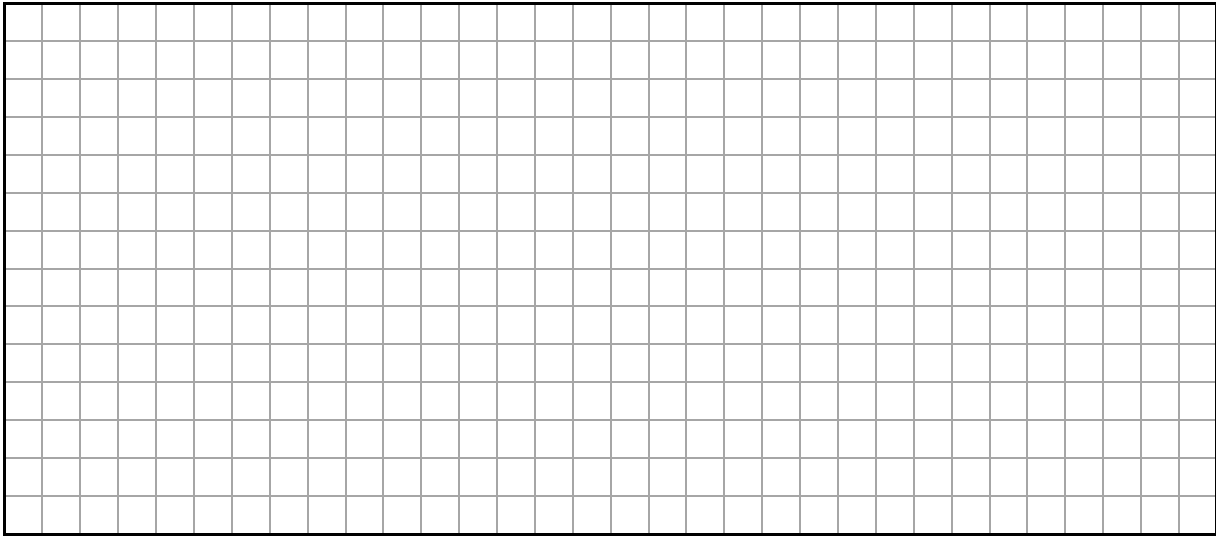
The co-ordinate diagram below shows a part of the town where Colm lives. Some of the favourite places Colm likes to go to with his friends, the beach (*B*), the sports field (*F*) and his home (*H*), are indicated in the diagram.



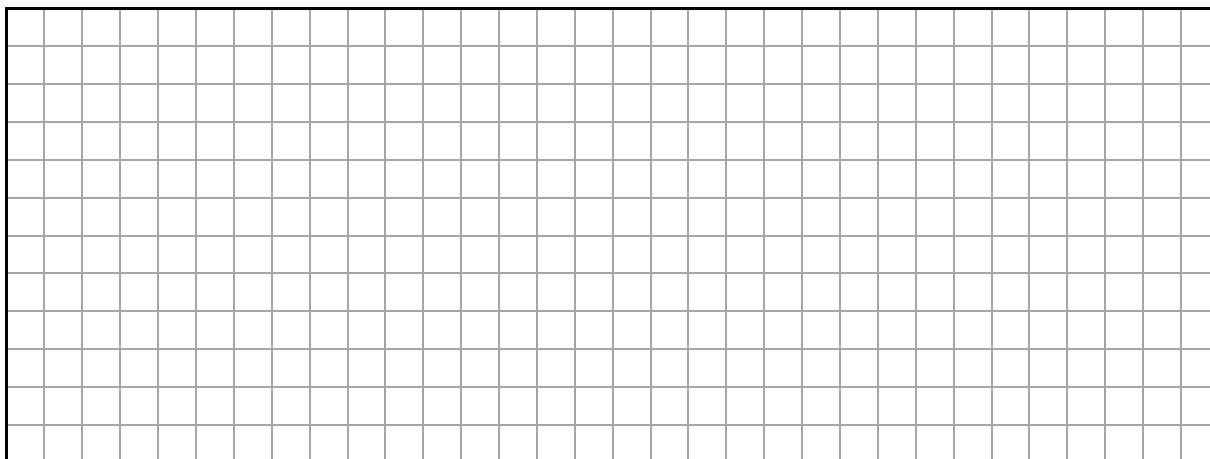
(a) Complete the table below to show the co-ordinates of Colm’s favourite places.

Point	<i>B</i>	<i>F</i>	<i>H</i>
Co-ordinates	(,)	(2, 4)	(,)

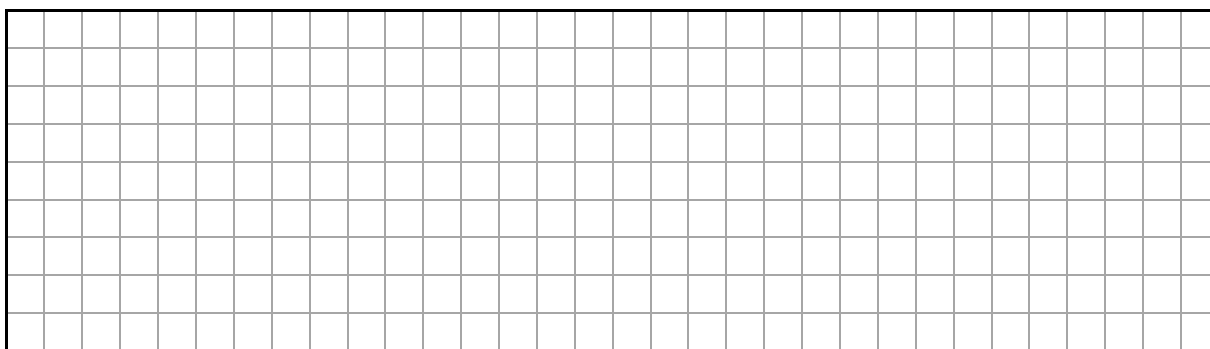
(b) Colm’s nearest shop is halfway between the beach (*B*) and his home (*H*).
Work out the co-ordinates of the shop, the **midpoint** of [*BH*].
Plot this point on the co-ordinate diagram above. Label this point *S*.



- (c) Each small square on the co-ordinate diagram has a side of length 1 cm.
Show that the **distance** between Colm's home (H) and the sports field (F) is 3.6 cm on the co-ordinate diagram when rounded to one decimal place.



- (d) The co-ordinate diagram is to a scale of 1 : 15 000.
Find the **actual distance** from Colm's home to the sports field.
Give your answer in metres.

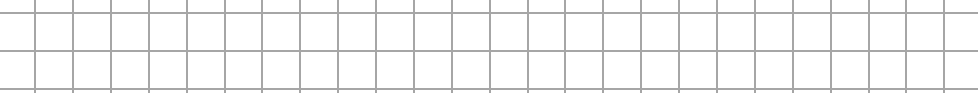


Question 3 (Suggested maximum time: 10 minutes)

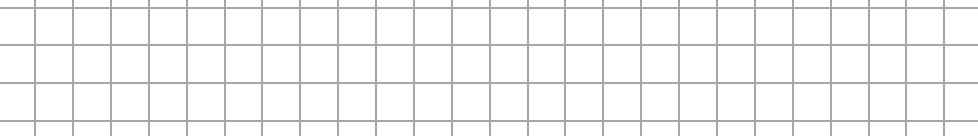
Question 3 (Suggested maximum time: 10 minutes)

- (a)** Simplify $3x + 4z + 9x - 2z$.

- (b)** Find the value of $12n - 6m$ where $n = 2$ and $m = 4$.



- (c)** Multiply out and simplify $(2p - 1)(3p + 6)$.



- (d)** Factorise $x^2 - 5x + 6$.

$$x^2 - 5x + 6 = (x \quad)(x \quad)$$



(Suggested maximum time: 5 minutes)

A is the set of prime numbers between 1 and 10.

(a) List the elements of the set A .

$$A = \{ \quad \quad \quad \}$$

$$B = \left\{ \begin{array}{c} \text{ } \end{array} \right\}$$

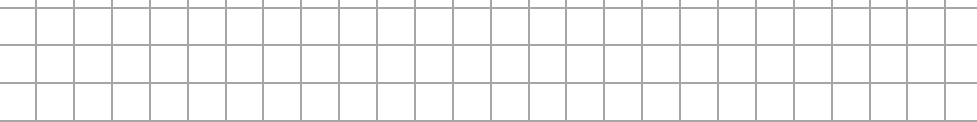
A Venn diagram illustrating the relationship between two sets, A and B , within a universal set U . The universal set U is represented by a large rectangle. Inside this rectangle, there are two overlapping ellipses. The left ellipse is labeled A and the right ellipse is labeled B . The two ellipses overlap in the center, representing the intersection of sets A and B .

$$A \cap B = \{ \quad \quad \quad \}$$
$$\#(A \cup B) = 6$$
[illegible]

Answer =

(Suggested maximum time: 10 minutes)

- , 6 , 9 , 12 ,



-

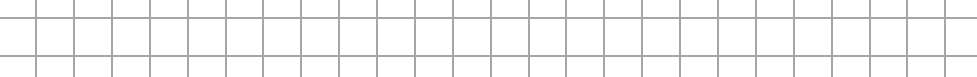
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-

11

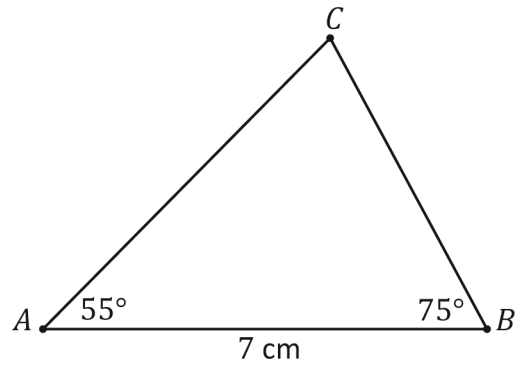
5

Justification:

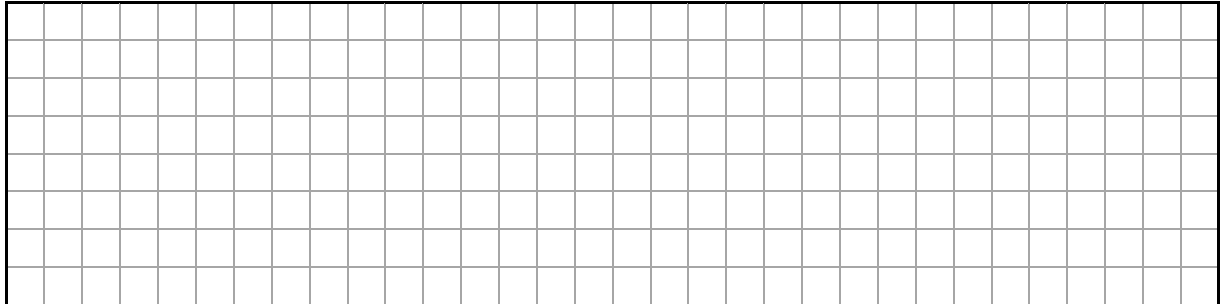
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Question 6**(Suggested maximum time: 10 minutes)**

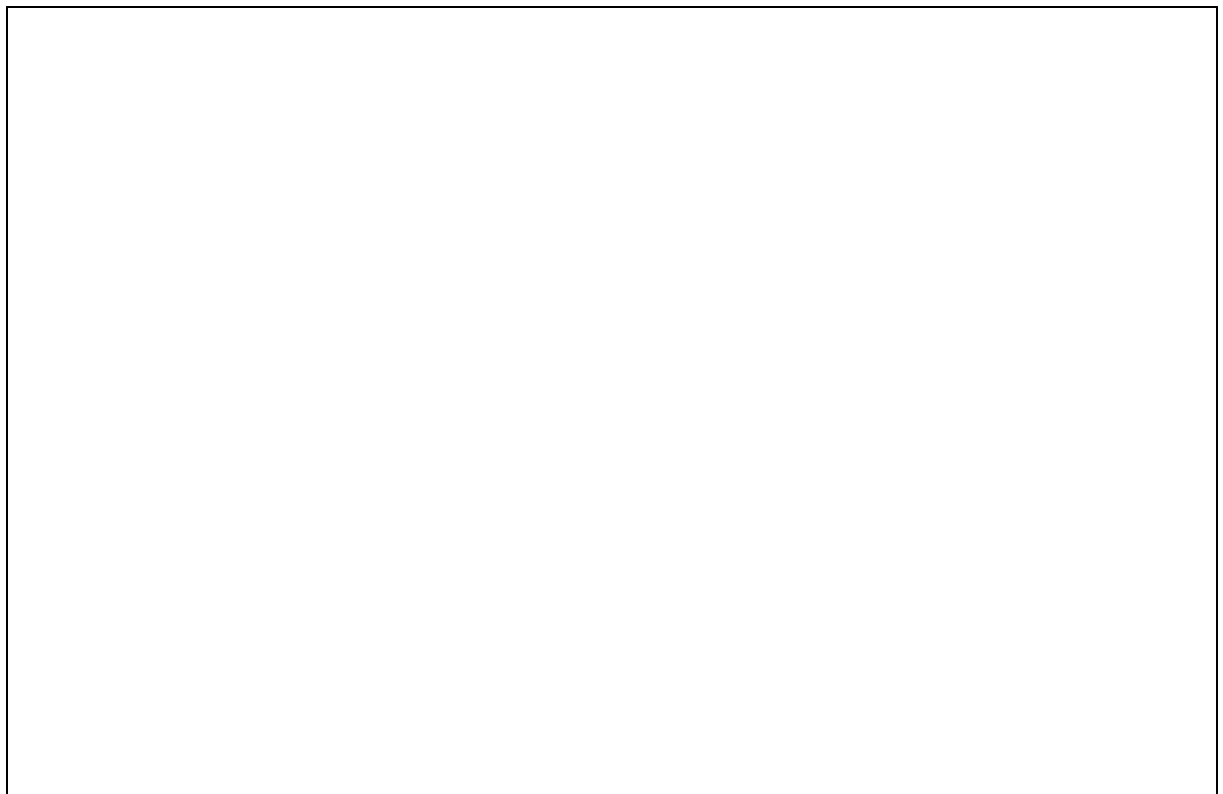
A triangle ABC has a side of length 7 cm and angles 55° and 75° , as shown in the diagram (not to scale).



- (a) Work out the size of the third angle in the triangle.



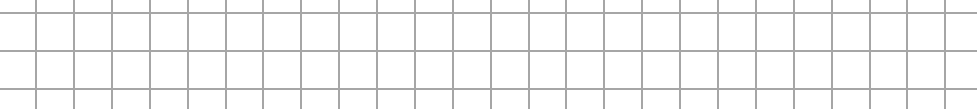
- (b) **Construct** the triangle ABC in the space below.
Show all of your construction lines clearly.



Question 7 (Suggested maximum time: 10 minutes)

A graphic showing a group of stylized human figures in various colors (red, orange, yellow, green, blue, purple) running in a line, suggesting a relay race or a team effort.

17	18	22	19	24
27	19	23	21	20
25	16	19	30	30

- 

- | | | | | | | | |
|---|--|--|--|--|--|--|--|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |

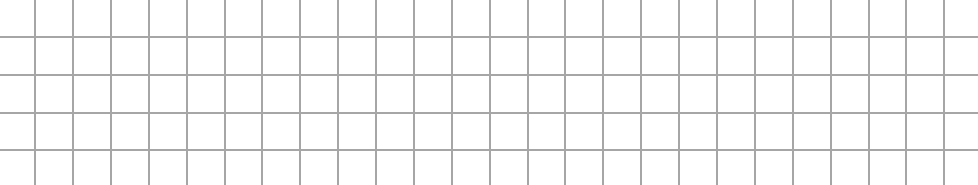
Mode =

- Median =

- (e) Work out the **mean** of the data.
Give your answer correct to the nearest minute.

[illegible]

- (f) It took Ellen 30 minutes to run the 10 km race.
Work out Ellen's **average speed** for the race, in km per hour.



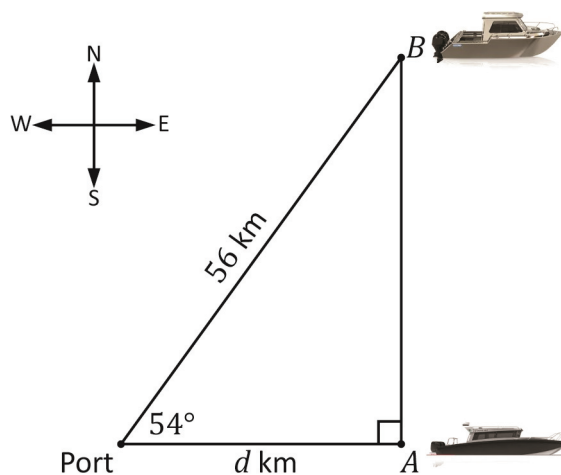
Question 8

(Suggested maximum time: 10 minutes)

Two boats are lined up at a port ready to depart.

Boat A sails due east and boat B sails 54° in a north-east direction.

When boat B has travelled 56 km and is directly north of boat A , as indicated in the diagram, boat A has travelled d km.



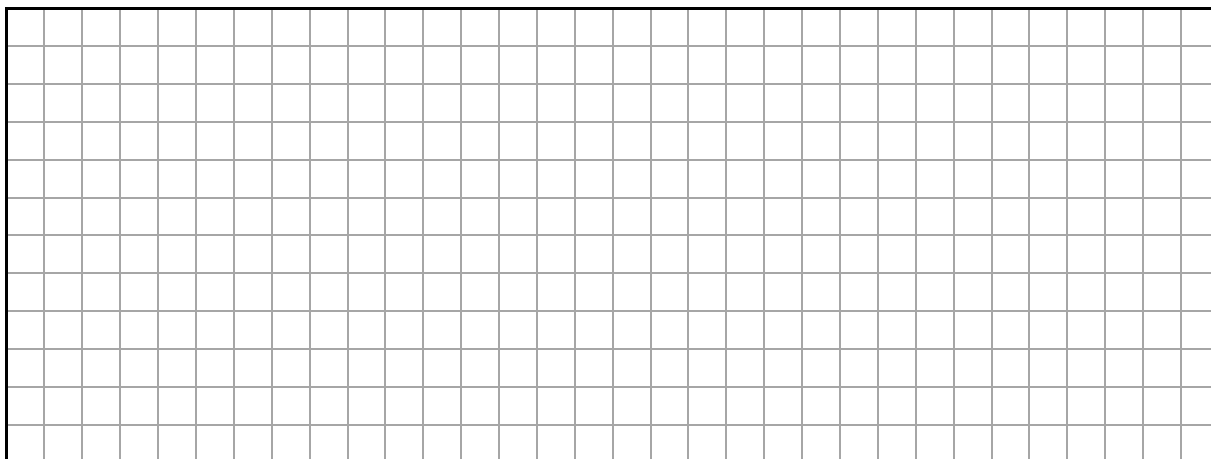
- (a) Use your calculator to find the value of $\cos 54^\circ$.
Give your answer correct to two decimal places.

$$\cos 54^\circ = \boxed{}$$

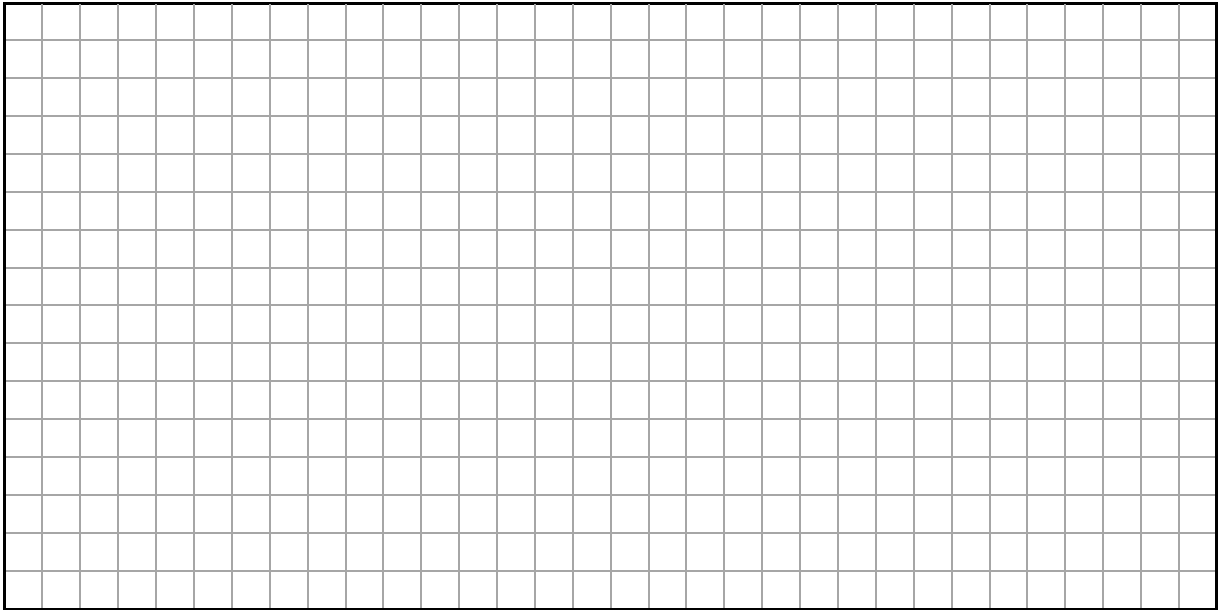
- (b) Use the right-angled triangle in the diagram to write $\cos 54^\circ$ as a fraction, in terms of d .

$$\cos 54^\circ = \frac{\boxed{}}{\boxed{}}$$

- (c) Use trigonometry to work out the value of d .
Give your answer correct to the nearest kilometre.



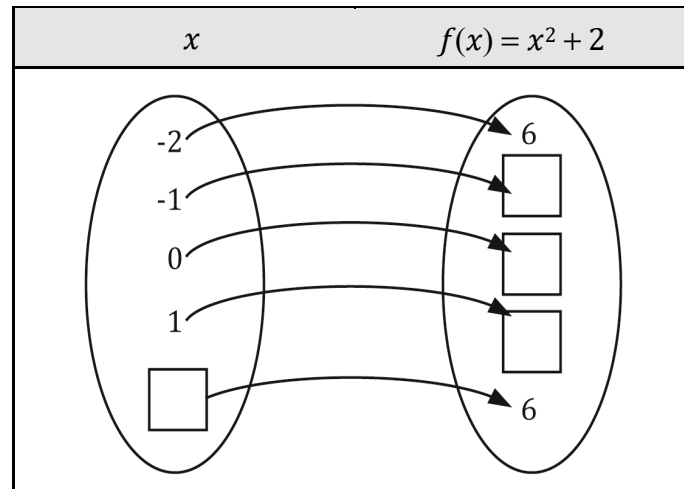
- (d) Using the theorem of Pythagoras, or otherwise, find the distance between boat *A* and boat *B*.
Give your answer correct to the nearest kilometre.



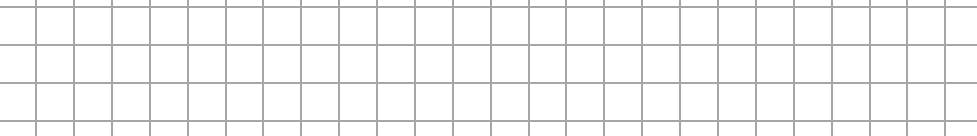
Question 9 (Suggested maximum time: 10 minutes)

Question 9 (Suggested maximum time: 10 minutes)

A mapping diagram of the function $f: x \rightarrow x^2 + 2$ is shown below.



- (a)** Fill in the four missing entries in the diagram.



- (b) (i)** Write down the **domain** of the function $f(x)$, as shown in the diagram.

Domain = $\{ \quad , \quad , \quad , \quad , \quad \}$

- (ii) Write down the **range** of the function $f(x)$, as shown in the diagram.

Range = { , , , }

Question 10

(Suggested maximum time: 10 minutes)

Martina is going on a holiday to Miami, Florida.
She books a flight from Dublin Airport to Miami.



- (a) The flight leaves Dublin at 11:10 am and arrives in Miami at 3:40 pm. All times are local.
- (i) Miami time is 5 hours behind Irish time.
What is the time Dublin when Martina lands in Miami?
Tick (✓) **one** box only.

10:40 am

☐

3:40 pm

8:40 pm

9

[illegible]

- (ii) How long is Martina's flight time to Miami?
Tick (✓) **one** box only.

8 hrs 30 mins

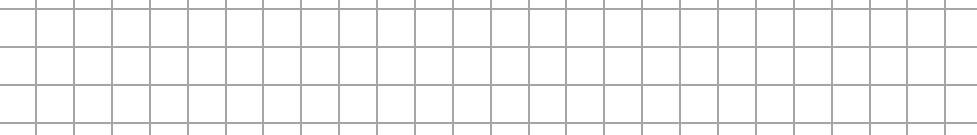
9 hrs 30 mins

10 hrs 30 mins

1



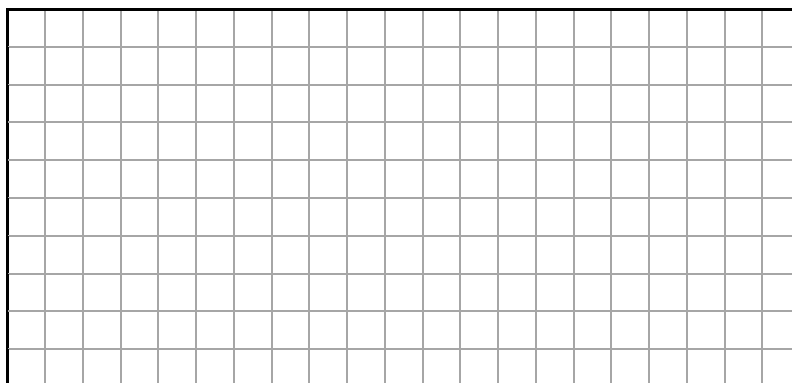
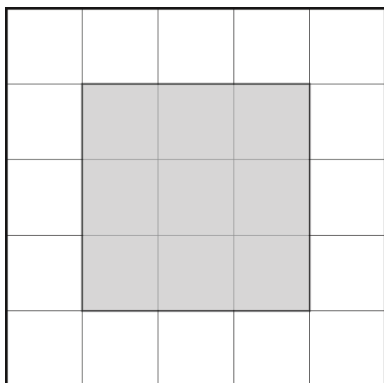
- (b)** Martina booked a hotel for her stay costing \$1440.
The currency conversion rate is €1 = \$1.09.
Convert \$1440 to euro (€). Give your answer correct to the nearest cent.



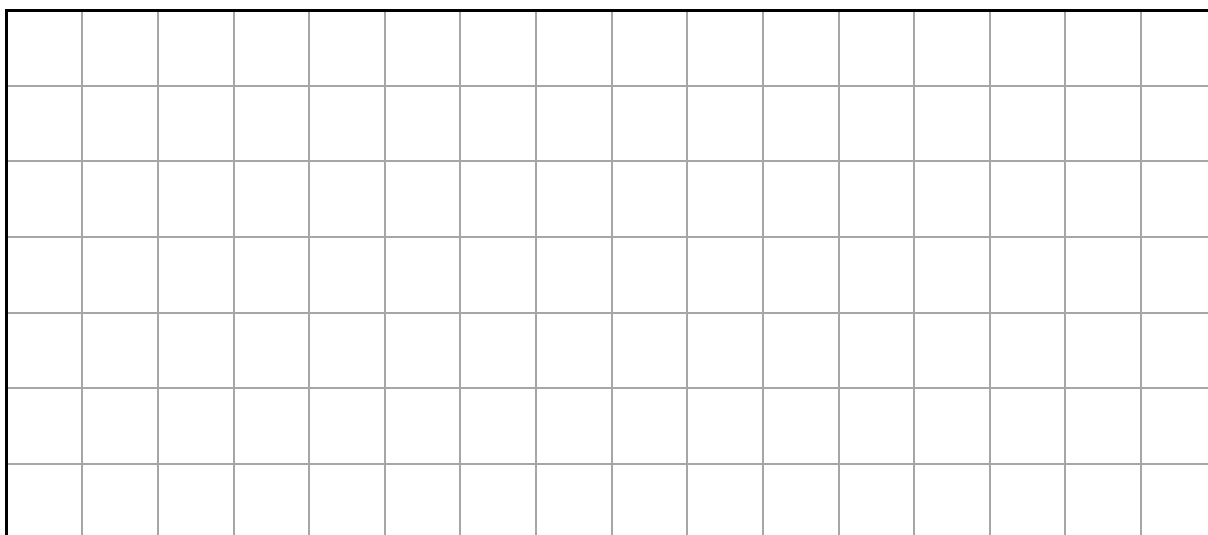
Question 11**(Suggested maximum time: 10 minutes)**

Each square in each of the grids below has sides of 1 cm in length.

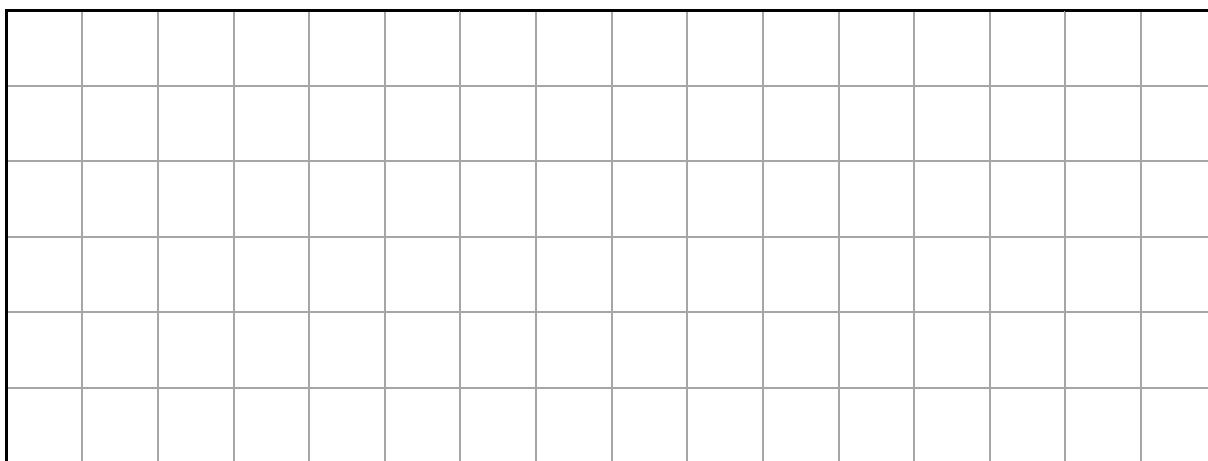
- (a)** Work out the area of the square that is shaded below.

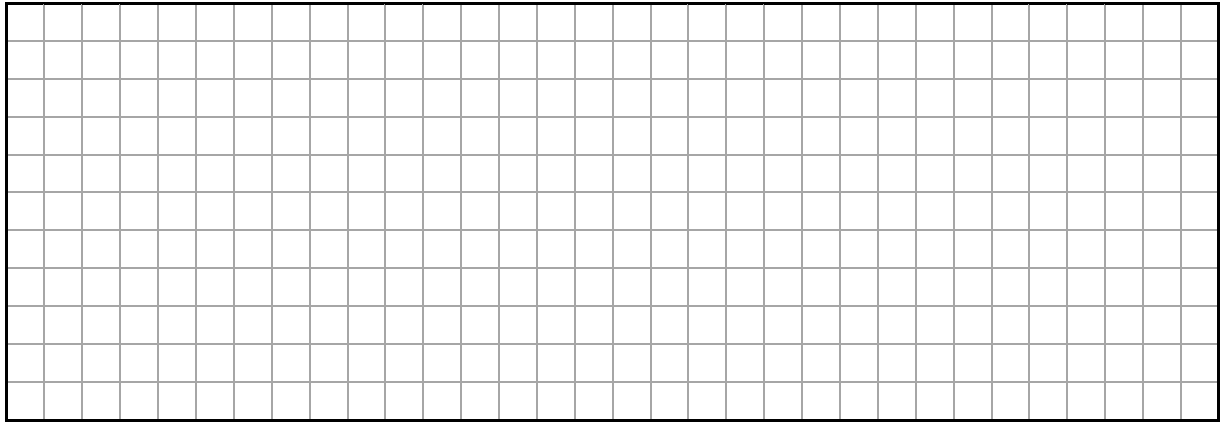


- (b)** A rectangle has an area of 20 cm^2 .
Construct the rectangle on the grid below.



- (c)** A triangle has an area of 6 cm^2 .
Construct the triangle on the grid below.





(d) The volume of a cube is 8 cm^3 .

(i) What is the length of one side of this cube?

Tick (✓) **one** box only. Give a reason for your answer.

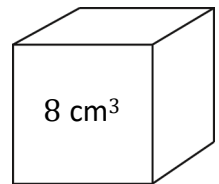
2 cm

☐

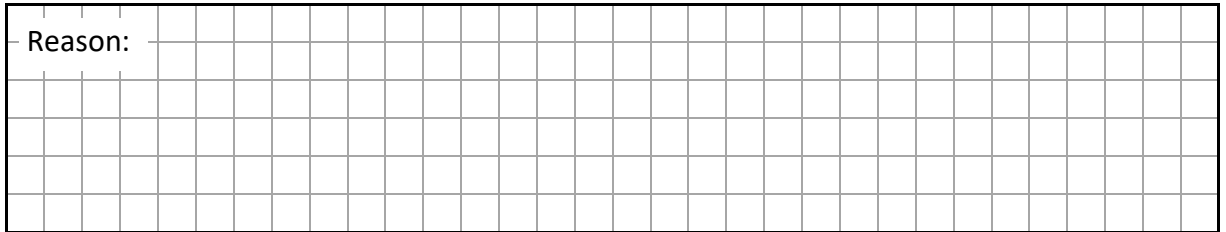
4 cm

☐

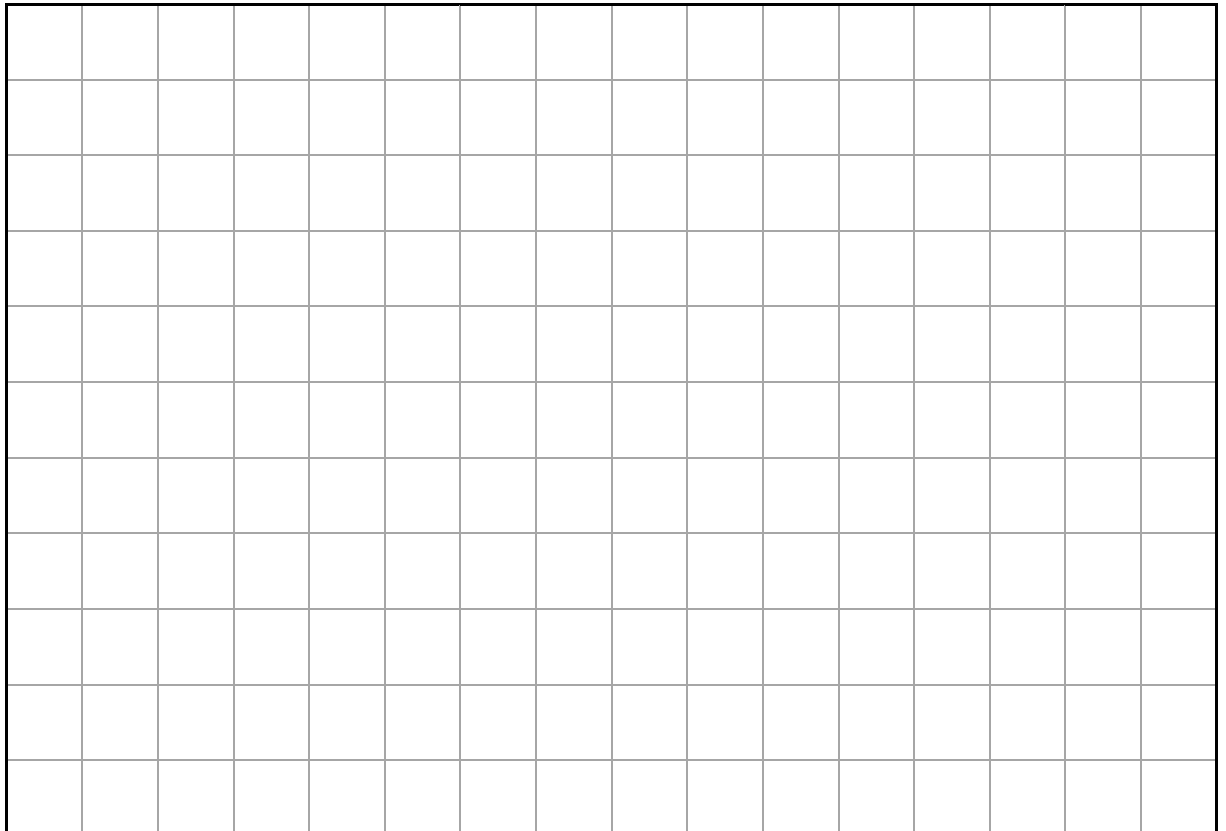
2.7 cm

☐


Reason:



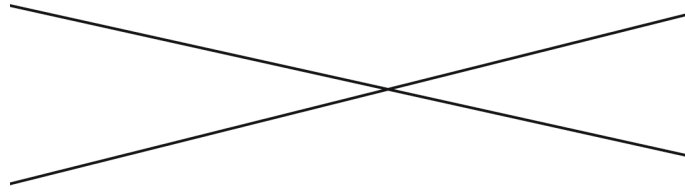
(ii) Draw a net of the cube, as accurately as you can.



Question 12

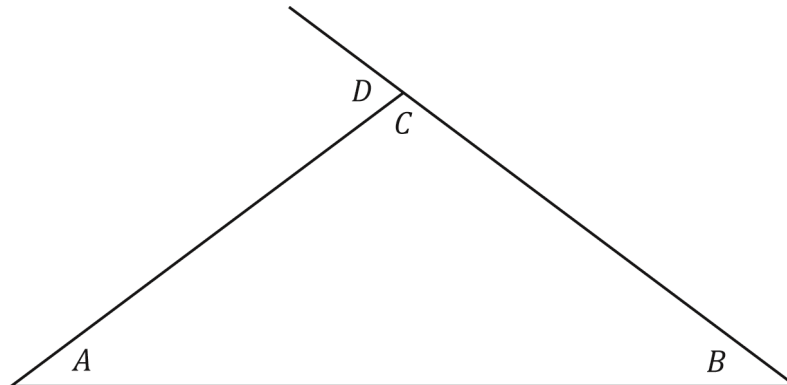
(Suggested maximum time: 5 minutes)

- (a)** The diagram shows two line segments.



- (i)** Write the letter **A** to indicate an **acute angle** on the diagram above.
- (ii)** Write the letter **O** to indicate an **obtuse angle** on the diagram above.

- (b)** A triangle has three interior angles A , B , C and one side extended to form angle D , as shown below.



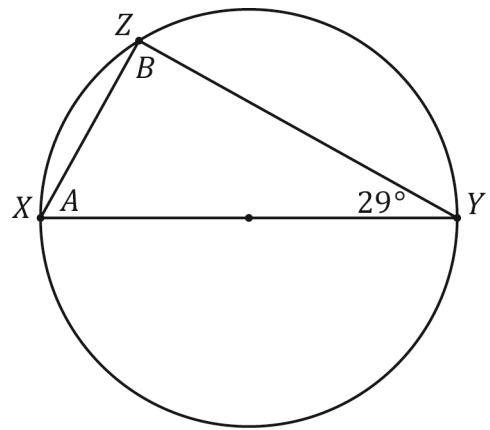
- (i)** Fill in the boxes to show three angles from the diagram above that sum to 180° .

$$\boxed{} + \boxed{} + \boxed{} = 180^\circ.$$

- (ii)** Fill in the boxes to show two angles from the diagram above that sum to 180° .

$$\boxed{} + \boxed{} = 180^\circ.$$

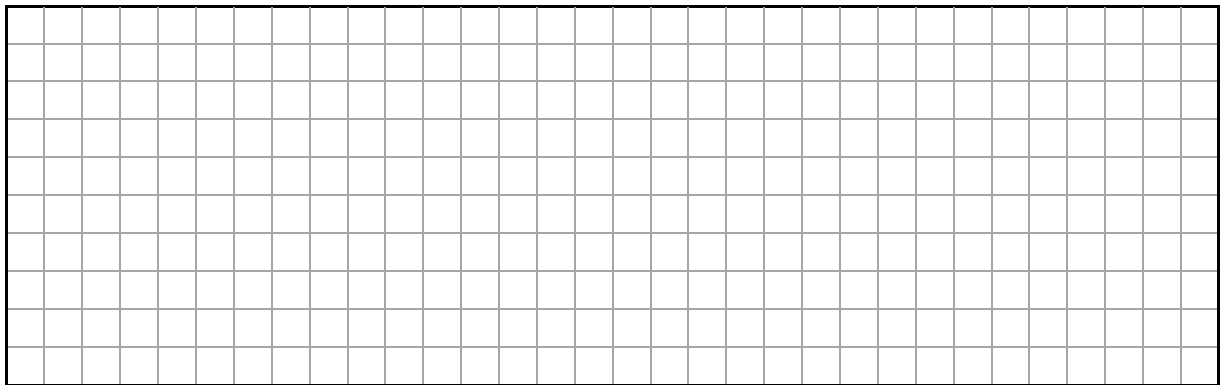
- (c) The diagram below shows a triangle standing on the diameter $[XY]$ of a circle. The point Z lies on the circle.



- (i) Write down the size of angle B in the triangle.

$|\angle B| =$

- (ii) Work out the size of angle A in the triangle.

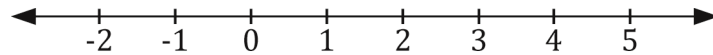


Question 13**(Suggested maximum time: 5 minutes)**

- (a) Graph the inequality $A < 3$ on the number line below, where $A \in \mathbb{N}$.

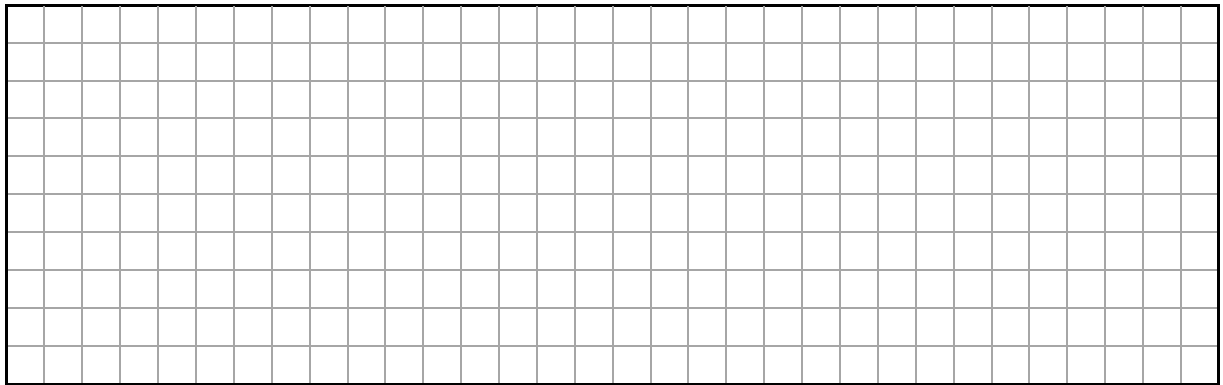


- (b) Graph the inequality $B < 3$ on the number line below, where $B \in \mathbb{R}$.



- (c) Solve the following inequality and show your solution on the number line below.

$$10C - 7 \leq 13 \text{ for } C \in \mathbb{Z}$$



Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Do not write on this page

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Pre-Junior Cycle Final Examination, 2024

Mathematics – Ordinary Level

Time: 2 hours

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A 2024-J117-1-EL-24/24